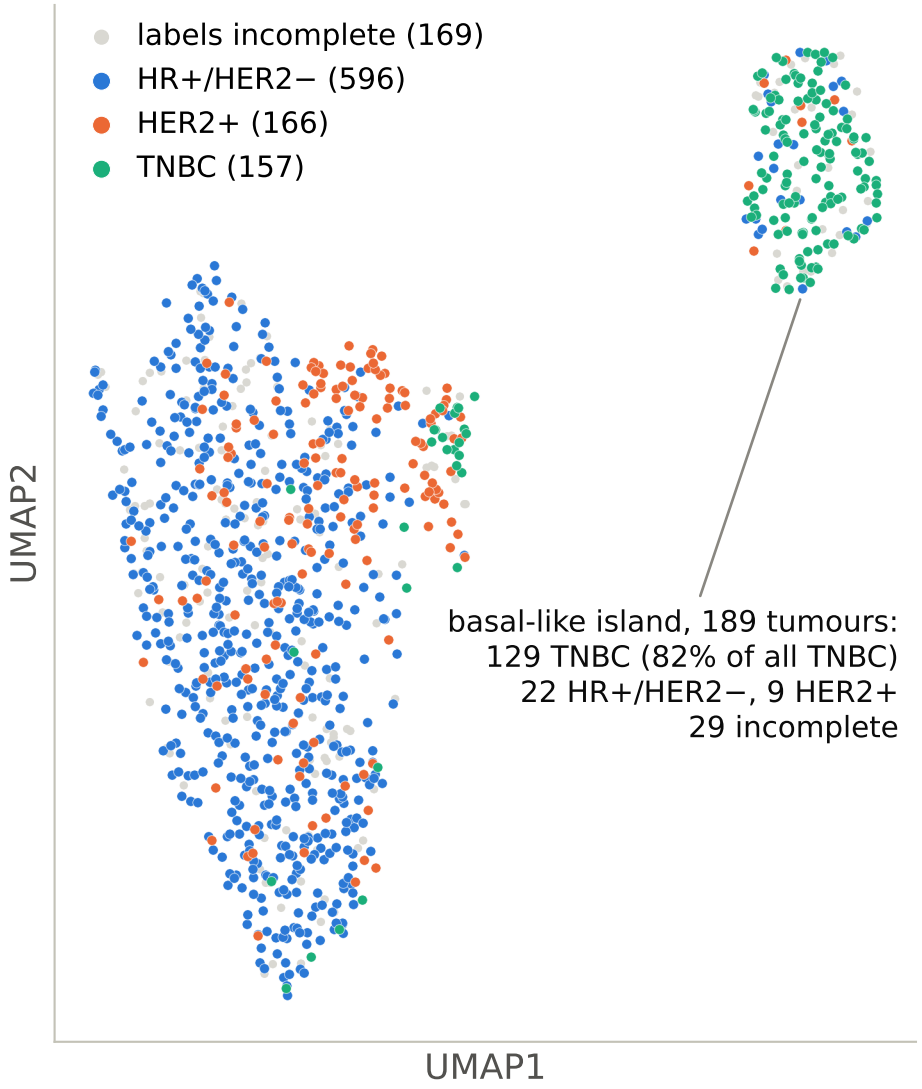
Expression sets TNBC apart: a basal-like island holds 82% of TNBC.

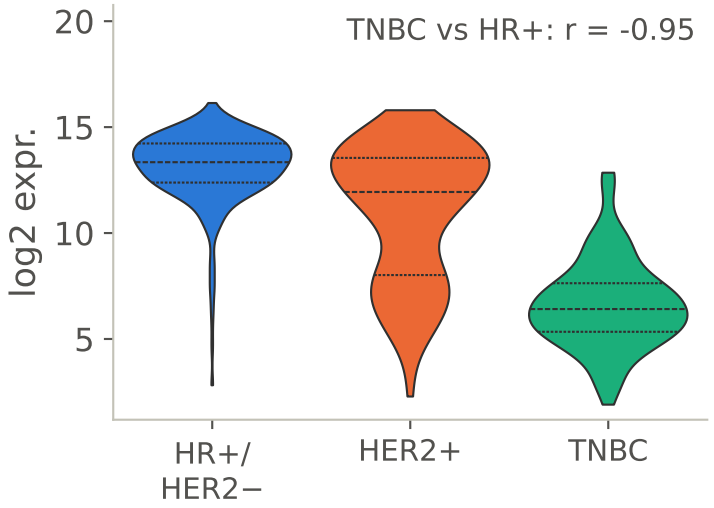
HER2+ tumours mostly overlap HR+/HER2− tumours; their clearest difference is ERBB2.

Violins: female supervised set (HR+/HER2− 591, HER2+ 160, TNBC 157); r = rank-biserial effect size; all four q < 1e-10 (Mann-Whitney, BH over 16 tests)

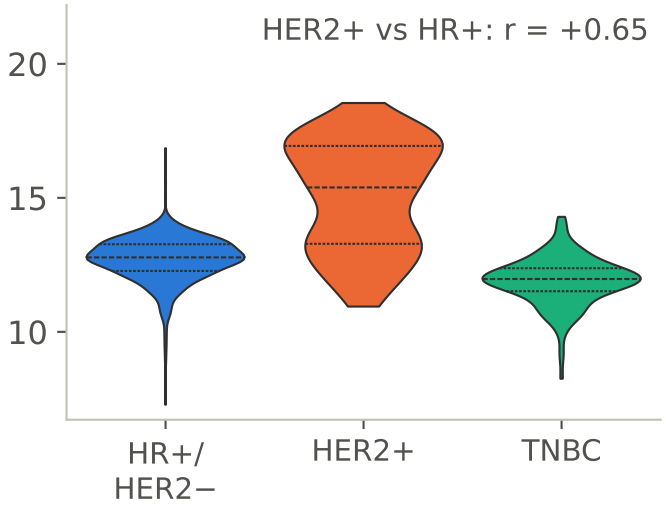
UMAP of 1,088 tumours (top 2,000 genes)



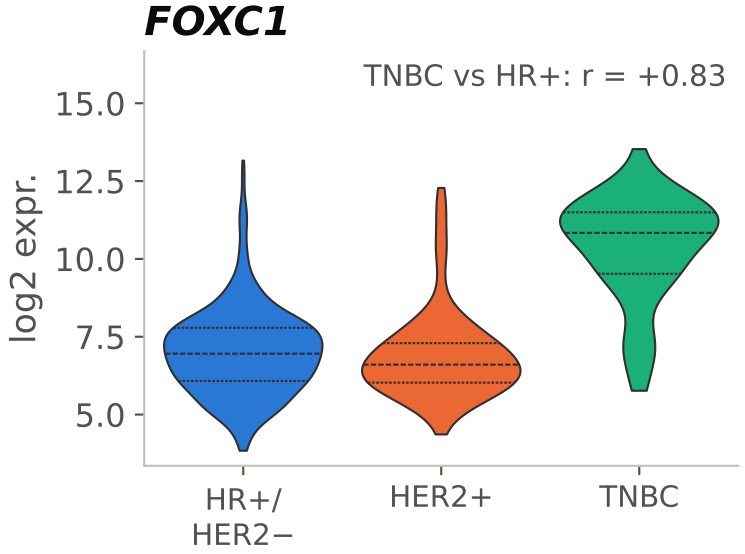
ESR1



ERBB2



FOXC1



MKI67

